

PAPER_955: BCS Phonon Resonance at ω_{SCm}

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Abstract

We analyze the BCS superconducting gap at the SCm phonon resonance frequency $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz. The gap equation $\Delta = (\hbar\omega_{\text{SCm}}/2) \tanh(\Delta/2k_B T) \cdot S_{26} \cdot (F_{UBi}/F_U)$ is solved self-consistently at the phonon resonance, revealing how the SCm vacuum phonon mode mediates Cooper pairing.

1. BCS Gap at Phonon Resonance

$$\Delta_{\text{res}} = \frac{\hbar\omega_{\text{SCm}}}{2} \cdot \tanh\left(\frac{\Delta}{2k_B T}\right) \cdot S_{26} \cdot \frac{F_{UBi}}{F_U}$$

2. Iterative Solution

Starting from $\Delta_0 = \hbar\omega_{\text{SCm}}/2$, iterate:

$$\Delta_{n+1} = \text{rhs}(\Delta_n, T)$$

until $|\Delta_{n+1} - \Delta_n| < \epsilon$.

3. Phonon Enhancement Factor

The resonance Q-factor at the phonon peak:

$$Q_{\text{res}} = \frac{\omega_{\text{SCm}} \cdot \sqrt{\Delta}}{k_B T}$$

4. Source Data

- **File:** et_phonon_resonance.py §6
 - **Session:** 214
 - **CP4 Class:** BCSPhononResonanceCalc (#539)
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References

1. Bardeen, Cooper, Schrieffer - Theory of Superconductivity (1957)
2. Murphy, D.T. - Star Magic UQFF Framework (2024-2026)